PHARMACOLOGY · MCQ PRACTICE

Drugs in Bowel Diseases

SGL 2
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. A medical student is asked which two conditions make up inflammatory bowel disease (IBD). What is the correct answer? EASY

A. Crohn's disease and ulcerative colitis
B. Irritable bowel syndrome and diverticulitis
C. Peptic ulcer disease and GERD
D. Celiac disease and lactose intolerance
E. Diverticulosis and appendicitis

2. A patient with ulcerative colitis is started on mesalamine. Which class of drug is this? EASY

A. A glucocorticoid
B. An aminosalicylate (5-ASA)
C. An immunomodulator
D. A biological response modifier
E. A JAK inhibitor

3. A patient is prescribed probiotics after a course of antibiotics. What is the main purpose of probiotics in this situation? EASY

A. To directly kill the causative bacteria responsible for the infection
B. To suppress the immune system so inflammation subsides more quickly
C. To restore gut flora and reduce the risk of Clostridium difficile infection
D. To neutralize stomach acid and protect the gastric mucosa from irritation and damage
E. To inhibit intestinal motility and reduce the frequency of bowel movements

4. A patient with IBS-D is prescribed loperamide. What type of drug is this? EASY

A. An antibiotic that reduces bacterial overgrowth in the gut
B. A selective serotonin reuptake inhibitor used for mood symptoms
C. An opioid-receptor agonist/antagonist controlled substance restricted to severe cases
D. An over-the-counter antidiarrheal that slows intestinal contractions
E. A bile acid binder that is taken with meals

5. A patient with Crohn's disease develops fistulae and strictures. What type of inflammation underlies these complications? MEDIUM

A. Continuous mucosal inflammation limited strictly to the colon and rectum
B. Inflammation limited to the rectum alone with sparing of the rest of the colon
C. Submucosal-only inflammation with no transmural extension whatsoever
D. Inflammation limited to the small intestine only, sparing the colon entirely
E. Patchy transmural inflammation that can affect any part of the GI tract

6. A patient with ulcerative colitis has been on long-term mesalamine therapy for extensive disease. What additional benefit does the lecture associate with long-term 5-ASA therapy? MEDIUM

A. It reduces the risk of colorectal cancer by about 75%
B. It cures the underlying disease permanently with no chance of relapse
C. It eliminates the need for any future colonoscopic surveillance entirely
D. It prevents all future relapses of the disease indefinitely without exception
E. It reverses existing colonic strictures that have already formed

7. A patient on budesonide for IBD asks why this glucocorticoid causes fewer systemic side effects than prednisolone. What property explains this? MEDIUM

A. Budesonide is not absorbed from the GI tract at all under any circumstances
B. Budesonide undergoes high first-pass metabolism
C. Budesonide binds a different receptor than other glucocorticoids such as prednisolone
D. Budesonide is given only as a rectal enema and never orally
E. Budesonide has a much shorter half-life than prednisolone in the circulation

8. A patient on azathioprine for Crohn's disease develops unexpectedly severe myelosuppression. Which enzyme's activity, when reduced by genetic polymorphism, increases this risk? MEDIUM

A. Dihydrofolate reductase in the liver
B. Cytochrome P450 3A4 in the gut wall
C. Thiopurine methyltransferase (TPMT)
D. Xanthine oxidase in uric acid metabolism
E. Thymidylate synthase in DNA synthesis

9. A patient with Crohn's disease is switched from azathioprine to methotrexate. Which statement about methotrexate in IBD is correct? MEDIUM

A. It is instead more effective in ulcerative colitis than in Crohn's disease
B. It has a rapid onset of action within 24 hours of the first dose
C. It works by directly binding TNF-α in the systemic circulation
D. It is more effective in Crohn's disease than in ulcerative colitis
E. It is only available as an oral formulation, never given parenterally

10. A patient on vedolizumab for refractory ulcerative colitis asks about its safety compared to other biologics like natalizumab. What does the lecture state about its risk profile? MEDIUM

A. It carries a substantially higher risk of PML than natalizumab does
B. It has no reported adverse effects of any kind in any patient population
C. It is contraindicated in all IBD patients due to an unacceptably high infection risk
D. It works identically to anti-TNF agents such as infliximab and adalimumab
E. It has a safer systemic profile, with a much lower risk of progressive multifocal leukoencephalopathy (PML) than natalizumab

11. A patient with ulcerative colitis is started on tofacitinib. What is this drug's mechanism of action? MEDIUM

A. It inhibits Janus kinases (mainly JAK1/3), blocking STAT signaling and cytokine-driven inflammation
B. It binds TNF-α directly in the systemic circulation
C. It blocks α4β7 integrin on circulating lymphocytes
D. It binds the p40 subunit shared by IL-12 and IL-23 receptors
E. It inhibits dihydrofolate reductase in lymphocytes

12. A patient on ustekinumab for moderate-to-severe Crohn's disease asks how the drug works. What is the correct mechanism? MEDIUM

A. It binds TNF-α, preventing its interaction with the TNF receptor
B. It binds the p40 subunit shared by IL-12 and IL-23, inhibiting Th1/Th17 pathways
C. It inhibits Janus kinases such as JAK1 and JAK3
D. It blocks lymphocyte trafficking via α4β7 integrin binding
E. It binds sphingosine-1-phosphate receptors on lymphocytes

13. A patient with moderate-to-severe ulcerative colitis is started on ozanimod. How does this drug reduce inflammation? MEDIUM

A. It binds TNF-α directly in the systemic circulation at high affinity
B. It inhibits Janus kinases such as JAK1 and JAK3 in lymphocytes
C. It modulates S1P1 and S1P5 receptors, causing lymphocytes to remain within lymph nodes and reducing their circulation
D. It blocks dihydrofolate reductase in rapidly dividing lymphocyte populations
E. It binds the p40 subunit of IL-12 and IL-23 on antigen-presenting cells

14. A patient with Crohn's disease is having a flare-up and asks about dietary management. According to the Crohn's disease exclusion diet (CDED) discussed in the lecture, which foods should be cut out? MEDIUM

A. Only red meat, with every other food group left unrestricted entirely
B. Fruits, vegetables, rice, and potatoes, which are in fact the encouraged foods
C. Only dairy products, with no other restrictions placed on any other food
D. Wheat and other gluten-containing grains, dairy products, animal fats, additives/ultra-processed foods, and red meat
E. Lean meats and eggs, which are in fact among the encouraged foods

15. A patient develops Stevens-Johnson syndrome while on 5-ASA therapy. Which agents in this class does the lecture specifically associate with headache and skin rashes including this reaction? HARD

A. Only sulphasalazine, with none of the other 5-ASA agents implicated
B. Only glucocorticoids such as prednisolone and budesonide
C. Only biological response modifiers such as infliximab and vedolizumab
D. Only JAK inhibitors such as tofacitinib and upadacitinib
E. Mesalamine, olsalazine, and balsalazide

16. A patient taking olsalazine for ulcerative colitis develops diarrhoea. How does olsalazine's structure explain both its mechanism and this common side effect? HARD

A. It is a dimer of two 5-ASA molecules linked by an azo bond hydrolysed by colonic bacteria, releasing 5-ASA in the colon
B. It is directly and completely absorbed intact in the stomach and metabolized entirely in the liver
C. It is a single 5-ASA molecule requiring no bacterial activation step of any kind in the colon
D. It is a prodrug activated by hepatic CYP3A4 enzymes rather than by colonic bacteria
E. It releases sulfapyridine rather than 5-ASA once it reaches the colon

17. A patient with steroid-dependent Crohn's disease is switched to a biologic but develops reactivation of latent hepatitis B. Which drug class is most associated with this specific risk? HARD

A. Anti-integrin agents such as vedolizumab, which have a safer systemic profile
B. Anti-TNF agents (e.g. infliximab, adalimumab)
C. JAK inhibitors such as tofacitinib and upadacitinib, taken orally
D. Aminosalicylates such as mesalamine, sulphasalazine, and olsalazine
E. S1P receptor modulators such as ozanimod, given orally once daily

18. A patient with IBS-D is prescribed eluxadoline. What is this drug's receptor mechanism, and why does it have limited central opioid effects? HARD

A. It is a pure μ-receptor antagonist with substantial, clinically significant CNS penetration
B. It is a serotonin 5-HT3 receptor antagonist with high oral bioavailability throughout the body
C. It is a mixed μ/κ opioid-receptor agonist and δ-receptor antagonist with low oral bioavailability
D. It is a guanylate cyclase-C agonist acting directly on the intestinal epithelial surface
E. It is an S1P receptor modulator with substantial, clinically significant CNS penetration

19. A patient with IBS-C is started on linaclotide. What is its mechanism of action, and what adverse effect is commonly recognised? HARD

A. It is a mixed opioid agonist and antagonist acting on enteric neurons in the gut wall; constipation is a recognised adverse effect of this mechanism
B. It blocks sodium absorption in the lining of the small intestine; hypertension is a recognised adverse effect of this mechanism
C. It is a serotonin 5-HT3 receptor antagonist acting centrally in the brainstem; QT prolongation is a recognised adverse effect of this mechanism
D. It agonizes guanylate cyclase-C on the intestinal epithelium, increasing cGMP and chloride/bicarbonate secretion; diarrhoea is a recognised adverse effect
E. It inhibits dihydrofolate reductase within rapidly dividing intestinal epithelial cells; hepatotoxicity is a recognised adverse effect

20. A critically ill ICU patient with a central venous catheter is being considered for probiotic supplementation. What does the lecture say about this situation? HARD

A. Probiotics are completely and unconditionally safe to use in all ICU patients, regardless of catheter status or illness severity

B. Probiotics are contraindicated only in patients who also happen to have lactose intolerance as a separate condition

C. Probiotics should be given at substantially higher doses in ICU patients specifically in order to help prevent infection

D. Probiotics are contraindicated only in patients who are currently being treated for an H. pylori infection

E. Probiotics should be used with caution in critically ill/ICU patients and those with central venous catheters, due to risk of sepsis

✓ Answer key

Q	Answer	Difficulty	Why
1	**A**	EASY	IBD includes Crohn disease (CD) and ulcerative colitis (UC).
2	**B**	EASY	Mesalamine (mesalazine) is 5-aminosalicylic acid (5-ASA), an aminosalicylate, alongside sulphasalazine, olsalazine, and balsalazide.
3	**C**	EASY	Probiotics help restore gut flora and reduce the risk of Clostridium difficile infection in antibiotic-associated diarrhea.
4	**D**	EASY	Loperamide is an OTC antidiarrheal that controls diarrhea and slows intestinal contractions; it is not proven to relieve abdominal pain and is typically for short-term use.
5	**E**	MEDIUM	Crohn's disease causes patchy transmural inflammation that can affect any part of the GI tract, leading to fistulae and strictures — unlike UC's continuous mucosal/submucosal injury limited to the colon and rectum.
6	**A**	MEDIUM	Long-term aminosalicylate therapy for extensive disease is associated with about a 75% reduction in colorectal cancer risk.
7	**B**	MEDIUM	Budesonide (oral) undergoes high first-pass hepatic metabolism, giving it fewer systemic effects than other oral glucocorticoids such as prednisolone.
8	**C**	MEDIUM	Thiopurines are metabolized by thiopurine methyltransferase (TPMT); genetic polymorphisms reducing TPMT activity increase toxicity risk, including myelosuppression.
9	**D**	MEDIUM	Methotrexate (given orally, IM, or SC weekly) inhibits dihydrofolate reductase and is more effective in Crohn's disease than in ulcerative colitis.
10	**E**	MEDIUM	Vedolizumab (an anti-α4β7 integrin antibody) blocks lymphocyte trafficking and has a safer systemic profile with a much lower PML risk than natalizumab.
11	**A**	MEDIUM	Tofacitinib (approved for UC) and upadacitinib (UC and Crohn's) are oral JAK inhibitors, mainly inhibiting JAK1/3 and blocking STAT signaling and cytokine-driven inflammation; adverse effects include infections (e.g. herpes zoster), hyperlipidemia, and cytopenias.
12	**B**	MEDIUM	Ustekinumab binds the p40 subunit shared by IL-12 and IL-23, inhibiting Th1/Th17 pathways; it is used for moderate-to-severe Crohn's disease and ulcerative colitis.
13	**C**	MEDIUM	Ozanimod is an oral S1P receptor modulator selective for S1P1 and S1P5; it causes lymphocytes to remain within lymph nodes, reducing their circulation to tissues and decreasing inflammation.
14	**D**	MEDIUM	The CDED recommends cutting out wheat/gluten-containing grains, dairy, animal fats, additives/ultra-processed foods, and red meat, while encouraging whole foods, fruits, vegetables, rice, potatoes, lean meats, and eggs.
15	**E**	HARD	Mesalamine, olsalazine, and balsalazide have a lower incidence of side effects than sulphasalazine overall, but may still cause headache and skin rashes such as Stevens-Johnson syndrome; diarrhoea is common specifically with olsalazine.
16	**A**	HARD	Olsalazine is a 5-ASA dimer linked by an azo bond that colonic bacteria cleave into two molecules of 5-ASA; it is poorly absorbed orally, and diarrhoea is a common side effect.
17	**B**	HARD	Anti-TNF agents carry adverse effects including infusion reactions, infections (especially TB and hepatitis B reactivation), demyelination, and lymphoma — a risk profile distinct from vedolizumab's safer systemic profile.
18	**C**	HARD	Eluxadoline is a mixed μ and κ opioid-receptor agonist and δ-receptor antagonist acting on enteric neurons; its low oral bioavailability limits adverse effects on central opioid receptors.
19	**D**	HARD	Linaclotide is a synthetic guanylin-like peptide that agonizes guanylate cyclase-C on the luminal intestinal epithelium, increasing cGMP and chloride/bicarbonate secretion and accelerating transit; diarrhoea is a recognised adverse effect.
20	**E**	HARD	Probiotics are generally safe, especially in healthy individuals, but should be used with caution in immunocompromised patients, critically ill/ICU patients, and those with central venous catheters (risk of sepsis).